

NOVA

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Information
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School

Data Mining Clustering Project Proposal Report

Amazing International Airlines Inc.

[Master's in Data Science and Advanced Analytics]



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1.0 Executive Summary:

This report details the transition of raw data to an actionable behavioural segmentation map. We engineered features to analyse customers' activity through three major lenses: **Customer Value**, **Flight Frequency**, and **Seasonality**. Our analysis confirmed **Seasonality** as the primary driver of customer behaviour. Consequently, we propose a **Calendar-Driven Marketing** to maximise economic yield and increase Customer Lifetime Value.

To ensure data integrity, appropriate preprocessing techniques were applied. Zero-values in income and flight activity were preserved as meaningful behavioural signals (e.g., "Solo status"), while physical impossibilities (e.g., flight distances >16,000km) and system errors were removed.

We employed multiple clustering algorithms, including Hierarchical Clustering, K-Means, SOM, DBSCAN, Meanshift, Gaussian Mixture Models, and SOM + K-Means across the three segments (**Value**, **Frequency**, and **Seasonality**). **Hierarchical Clustering** applied to Seasonal Features emerged as the superior model. It provided the most actionable separation of the customer base, proving that **when** customers fly is a stronger differentiator than **how much** they spend.

The results of our analysis partition the customer base into four distinct groups:

Group	Market Share	Profile	Strategic Recommendations
Routine Travelers	69%	Year-round consistency; highest tenure and flight volume.	Drive retention through operational excellence and targeted, high-impact communication to solidify brand connection.
Summer Vacationers	21.7%	High intensity in summer; highest points redemption rate; shorter distances.	Shift marketing to March/April to capture early booking intent and market add-ons (baggage, insurance) for family travel.
Autumn Escapees	5.7%	Active almost exclusively in autumn; minimal year-round engagement.	Activate price-led "City Breaks" (low-fare, short-stay trips) promotions in Oct/Nov to stimulate demand and fill capacity in low-demand periods.
Winter Elite	3.7%	Premium long-haul travellers; highest average distance; travel with companions.	Deploy high-value offers for exotic destinations in Oct/Nov, focusing on premium service over price discounting

Results stipulate that Seasonality is the primary behavioural driver, partitioning the customer base into 31% high-value seasonal niches and 69% routine travellers. This distinction enables optimised, calendar-based marketing. We propose that further initiatives that prioritise micro-segmentation to isolate price versus convenience sensitivity, alongside predictive AI to forecast churn.

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2.0 Introduction

Our preliminary findings in Phase 1 transition the project from raw data to behavioural/pattern segmentation. Based on the EDA results, the engineered features enabled the grouping of customers across the following segments:

1. **Value:** enables the quantifying of customers' economic worth
2. **Traveller's Behaviour:** describes the travel patterns of customers. However, to better extrapolate more refined results, we split this segment into **Seasonal Patterns** and **Frequency and Volume Patterns**.
3. **Loyalty:** Measuring customers' engagement beyond flights. However, to better extrapolate more refined results, we decided to **drop** this segment and not utilise it for the clustering.

These engineered features transform the dataset into a multidimensional behavioural map. By applying unsupervised learning (clustering) techniques, we aim to segment customers based on three distinct dimensions: **Seasonal Preferences** (when they fly), **Flight Behaviour** (how much they fly), and **Customer Value** (financial contribution).

2.1 Preprocessing

2.1.1 Missing Values & Data Integrity

1. **Missing Data Analysis** An initial review showed that the flights dataset was complete, with no missing values. The customer dataset, however, required attention in three specific areas:
 - a. **Income & Customer Lifetime Value (20 records):** 20 customers had **missing income** and **lifetime value data**, and **no recorded flight history**. As these records provided no meaningful insight, they were removed to maintain data quality.
 - b. **Cancellation Date (14,613 records):** We identified **14,613** records with **no recorded cancellation date**. These customers were therefore classified as **active**.
2. **Zero (0) Values vs. Missing Data:** It is important to distinguish between missing data (NaN) and zero values (0), as the zeros (0) in this dataset represent meaningful business realities:
 - a. **Income (\$0):** Values of zero (0) in the **Income** feature were treated as valid data points rather than errors. They may likely represent specific demographic segments such as students, retirees, or unemployed individuals.
 - b. **Flight Activity:** Zero (0) values in features like **NumFlights**, **DistanceKM**, and **NumFlightsWithCompanions** are expected. They accurately describe customers who signed up but haven't flown (inactive), or customers who fly exclusively solo (for zero (0) values in the **NumFlightsWithCompanions** feature).
 - c. **Loyalty Points:** Zeros in **PointsAccumulated** or **PointsRedeemed** simply indicate customers who have not yet engaged with the loyalty program's reward system.

Conclusion: These zero values are not data gaps but rather distinct behavioural signals (e.g., "Inactive" or "Solo Traveler") that are essential for our clustering algorithms.

2.1.2 Duplicates

We identified duplicates in **1.9%** of the dataset. Since these were **random** errors with no strategic value, we **removed** them to ensure accuracy.

Commented [C02]: Establish context by connecting Phase 1 findings to clustering objectives. Document data preparation decisions and feature engineering based on Phase 1 findings.

2.1.3 Handling Incoherences

1. **Removal of Inactive Profiles:** We excluded all customers identified as "**Never Flown**" (8.9% of the dataset).
2. **Correction of Flight Data:** For records showing **NumFlights** = 0 but **DistanceKM** > 0, we imputed the missing flight counts by dividing the recorded distance by the **global average distance per flight**, preserving the customer's travel history.
3. **NetPoints Outlier Filtration:** Customers with NetPoints below **-5,000** were **removed**, as such **extreme** negative balances are most likely data **errors** rather than valid activity.
4. **Retention of Temporal Inconsistencies:** We retained flights recorded **before** enrollment or **after** cancellation, likely reflecting **re-enrollment** or **data migration**, to avoid losing over **4,000** valid behavioural profiles.

2.1.4 Outlier Detection & Handling

We applied business logic to distinguish between actual data errors and "extreme" customer behaviours that indicate high value.

1. **Physical Impossibilities (Flight Distance):** We removed the **AvgDistancePerFlight** listing distances over **16,000 km**. Since no commercial aircraft can fly this far in a **single leg**, values exceeding this limit (some reached **30,000 km**) (see [Figure 1](#)) were classified as data entry errors and excluded.
2. **System Errors (Extreme Negative NetPoints):** While small negative point balances are normal due to **timing differences** or **promotions**, we identified massive deficits (e.g., **-20,000 points**) (see [Figure 2](#)) that are functionally impossible. We set a realistic cap at **-5,000** points to filter out these system glitches while keeping legitimate active users.
3. **Retention of Behavioural Outlier:** Crucially, we **kept** all other **extreme values** regarding income or flight frequency. In a loyalty program context, these "statistical outliers" are **not errors**—they are our VIPs. Retaining them is **essential** to identifying our most profitable customer segments.

2.1.5 Feature Engineering

Here, we have created new features to capture customer status, value, and behaviour. They include the following:

1. The status and value features are '**IsActive**', and **DaysActive**, **HasFlown** and **IsValuable**
2. To understand customers' interaction with the airline, we developed these **behavioural features**. **RedemptionRate**, **CompanionRatio**, and **AvgDistancePerFlight**.
3. Given the data contained Canadian customers, we developed **seasonal features** using the Northern Hemisphere seasons. These included four seasonal ratios (**SummerFlightsRaCompanion RatioghtsRatio**, **SpringFlightsRatio**, and **AutumnFlightsRatio**) as well as a **HolidayFlightsRatio** to capture flight activity during key Canadian holiday months.
4. Finally, based on seasonal features, we segmented customers by their travel behaviour into these feature types: **Business Traveller**, **Vacation Traveller**, and **Regular Traveller**

Note: These and other features with respective descriptions are detailed in **Table 1** in the Annexe.

3.0 Methodology

3.1 Feature Selection

3.1.1 Removed/Dropped Features

To ensure the clustering focused on meaningful **behaviour** rather than static or redundant attributes, we applied feature selection to the original and engineered data, excluding variables in the following categories:

1. **Demographic:** Features such as **Gender**, **Marital Status**, **Education**, **City**, **Country**, and **Postal Code** were removed. Our objective was to segment customers based on how they fly and what they spend, not who they are.
2. **Redundant & Collinear Features:** We removed absolute totals such as **TotalDistanceKM**, **TotalFlights** and **TotalPointsAccumulated** in favour of relative metrics like **AvgDistancePerFlight**, **AvgFlightsPerMonth** and **RedemptionRate**. The following reasons are justifications for dropping these features:
 - a. **Behavioural Distinction:** Absolute sums mask distinct behaviours; for instance, high-frequency short trips can result in the same total distance as a single long-haul flight. To avoid this, we use **averages** and **ratios**, ensuring the model separates users by **frequency** and **flight type**, capturing lifestyle nuances rather than aggregate volume.
 - b. **Redundancy:** Since our model includes **AvgDistancePerFlight**, **AvgFlightsPerMonth**, and **DaysActive**, the total volume is implicitly represented by the interaction of these variables. Furthermore, we specifically excluded **TotalPointsAccumulated** due to its **perfect 1:1 linear correlation** with **TotalDistanceKM**. Including perfectly correlated variables introduces severe multicollinearity, causing the algorithm to overweight the "Volume" dimension.
3. **Administrative & Date Attributes:** Raw dates (**EnrollmentDateOpening**, **CancellationDate**) were excluded as they are not mathematically suitable for distance-based clustering. Instead, we utilised the derived feature **DaysActive** to capture the temporal dimension of tenure.
4. **Pre-Defined Segmentations (Data Leakage):** We removed categorical labels created during the Exploratory Data Analysis phase, such as **LoyaltyProfile**, **DistanceProfile**, **TravelerType**, and **MainSeason**. Including these would cause "data leakage," as the clustering algorithm would simply reproduce the rules we manually created rather than discovering new, organic patterns.

3.1.2 Selected Features & Segmentation

To ensure **clear** and **actionable** insights, the analysis separates **Customer Value** from **Passenger Behavior**.

Crucially, we distinguish **Volume** from **Timing**. This ensures that *how often* customers fly does not hide *when* they prefer to travel, defining three core segments:

1. **Value:** This addresses financial impact, identifying profitability regardless of current activity levels. Features in this segment are **Income**, **Customer Lifetime Value**, **NetPoints** and **RedemptionRate**.
2. **Frequency & Volume:** Captures the customer's flight activity and the "style" of travel. Features in this segment are **DaysActive**, **AvgFlightsPerMonth**, **AvgDistancePerFlight**, and **CompanionRatio**.
3. **Seasonal Patterns:** Identifies customers' periodic preferences. Features in this segment are **SummerFlightRatio**, **WinterFlightRatio**, **SpringFlightRatio**, **AutumnFlightRatio**, and **HolidayFlightRatio**.

Commented [C03]: Present feature selection process, algorithm selection process, parameter tuning methodology, and implementation approach comparing multiple clustering methods.

3.1.3 Feature Normalisation

To prepare the data for distance-based algorithms (like K-Means), we applied **Log-Transformation** to highly skewed monetary variables (**Income**, **Customer Lifetime Value**, **TotalPointsRedeemed**). Subsequently, we applied **Standard Scaling** to centre the data, ensuring that all features **contributed equally to the distance calculations** regardless of their original magnitude.

3.2 Clustering Perspectives

To reflect the different dimensions of passenger demand, we applied separate clustering to the **scaled data** for each segment. Multiple algorithms were tested and the most suitable was selected based on performance metrics (**R^2** , **Silhouette**, and **Davies-Bouldin scores**) and the ability to form **balanced, meaningful** groups. Cluster quality was validated through profiling **heatmaps**, **label distribution**, and visual separation using **UMAP**. This section outlines the selected clustering methods.

3.2.1 Value Feature Clustering

3.2.1.1 Hierarchical Clustering

To determine the optimal segmentation structure, we analysed dendrograms and performed a systematic grid search across multiple configurations (Linkages: Ward, Complete, Average; Metrics: Euclidean, Manhattan, Cosine) for various cluster counts. Based on the evaluation of metrics, the **Average Linkage with L1 Distance (k=4)** emerged as the superior model, offering the most balanced and cohesive clusters.

3.2.1.2 K-Means clustering

In our analysis, we assessed **K-means clustering** by first visualising the dispersion (inertia) across 1 to 19 clusters to identify the potential elbow point. We can observe in this figure that the elbow point is clearly **k=4** (see Figure 4). To refine this selection, we evaluated the model's performance metrics for cluster counts ranging from 2 to 7. We observed that the most robust segmentation was achieved when **k=4**.

3.2.1.3 SOM clustering

In our **Self-Organising Maps (SOM)** analysis, we first assessed training stability by monitoring the **Quantisation Error** curve over **2,000** iterations. We observed that model convergence stabilised between **900** and **1,200** iterations (see Figure 7). Based on this window, we conducted a hyperparameter optimisation search, testing variations in learning rate (**0.8, 0.9, 1**), and iteration counts.

The final model was configured on a **15 X 15 grid** with a learning rate of **1.0** and **1,020 iterations**. This resulted in **128 clusters**, a number deemed **too high** to be actionable for high-level customer profiling

3.2.1.4 Density-Based Clustering

3.2.1.4.1 Mean-Shift Clustering

In our analysis using **Mean Shift**, a density-based algorithm that does not require specifying the number of clusters a priori, the critical hyperparameter was the bandwidth (kernel size). To determine the optimal window for density estimation, we performed a sensitivity analysis by iterating through quantiles ranging from **0.05** to **0.70** using the `estimate_bandwidth` function.

Commented [C04]: 1st talk about the segmentation we did, of each of the perspective variables
2nd go into each perspective and talk about each clustering method (like last year's)
3rd for each perspective, choose the best model and explain
4th Final Cluster Discussion
5th CLUSTER ANALYSIS AND MARKETING APPROACHE
6th Assessing Feature Importance and Reclassifying the outlier
7th CONCLUSION

under is some idea to work on, not the final words, change it

Commented [C05]: 1. Which clustering method(s) produced the most interpretable and stable results for each perspective?
2. How many clusters best represent the customer base overall, and what evidence supports this decision
3. What differentiating features emerged as most important within each perspective, and how do they complement one another in the merged solution?
4. How would you describe the final set of customer segments

The most robust segmentation was identified at a **quantile of 0.2** (bandwidth of approx. **0.37**), resulting in a **2-cluster** solution

3.2.1.4.2 DBSCAN

For the **DBSCAN** analysis, we determined the optimal neighbourhood radius (eps) using the **K-distance graph** method with 20 nearest neighbours. Observing the 'knee' of the curve, we identified a target range between **0.4** and **0.6** (see Figure 11). We conducted a sensitivity analysis within this interval, testing different computational algorithms (ball_tree, kd_tree, brute), which resulted in identical results.

The final model was instantiated with **eps = 0.56**, identifying **3** distinct clusters (excluding outliers).

3.2.1.4.3 Gaussian Mixture Model

In our **Gaussian Mixture Models (GMM)** analysis, we initially determined the optimal number of components by minimising the **BIC** and **AIC** across a range of 1 to 10 components. The statistical 'elbow point' suggested **3** clusters (see Figure 13). To refine the model structure, we conducted a grid search evaluating different covariance types ('full', 'tied', 'diag', 'spherical') for cluster counts between 2 and 6. Although 'elbow point' suggested **3** clusters, the **5-cluster** solution using a '**spherical**' covariance type offered distinct advantages in terms of interpretability and business applicability and better metrics.

3.2.1.5 Model Combination - SOM + K-Means

We implemented a hybrid **SOM + K-Means** model to leverage the strengths of both algorithms. By applying K-Means to the features extracted from the SOM, we identified **3 distinct clusters**.

3.2.1.6 Model Assessment

We selected **K-Means (k=4)** as the final model due to its superior business interpretability and alignment with the airline's operational structure.

While other models produced competitive statistical scores, K-Means successfully identified four distinct and actionable personas: **Loyal High Spenders**, **Active Redeemers**, **Passive VIPs**, and **Average Users**.

3.2.2 Frequency & Volume Feature Clustering

3.2.2.1 Hierarchical Clustering

To determine the optimal segmentation structure, we analyzed dendrograms and performed a systematic grid search across multiple configurations (Linkages: Ward, Complete, Average; Metrics: Euclidean, Manhattan, Cosine) for various cluster counts.

Based on the evaluation of metrics, the **Average Linkage with Cosine Distance (k=3)** emerged as the superior model, offering the most balanced and cohesive clusters.

3.2.2.2 K-Means clustering

In our analysis, we assessed **K-means clustering** by first visualizing the dispersion (inertia) across a range of 1 to 19 clusters to identify the potential elbow point. We can observe in this figure that the elbow point is clearly **k=4** (see Figure 16). To refine this selection, we evaluated the model performance metrics for cluster counts between 2 and 8. We observed that the most robust segmentation was achieved when **k=4**.

3.2.2.3 SOM clustering

In our **Self-Organising Maps (SOM)** analysis, we first assessed training stability by monitoring the **quantisation Error** curve over **1,000** iterations. We observed that model convergence stabilized between **650** and **750** iterations ([see Figure 18](#)). Based on this window, we conducted a hyperparameter optimization search, testing variations in learning rate (**0.8, 0.9, 1**), and iteration counts.

The final model was configured on a **15 X 15 grid** with a learning rate of **1** and **730 iterations**. This resulted in **225 clusters**, a number deemed **too high** to be actionable for high-level customer profiling.

3.2.2.4 Density-Based Clustering

3.2.2.4.1 Mean-Shift Clustering

In our analysis using **Mean Shift**, a density-based algorithm that does not require specifying the number of clusters a priori, the critical hyperparameter was the bandwidth (kernel size). To determine the optimal window for density estimation, we performed a sensitivity analysis by iterating through quantiles ranging from **0.3 to 0.8** using the `estimate_bandwidth` function. The most robust segmentation was identified at a **quantile of 0.7** (bandwidth of approx. **0.28**) resulting in a **6-cluster** solution.

3.2.2.4.2 DBSCAN

For the **DBSCAN** analysis, we determined the optimal neighborhood radius (`eps`) using the **K-distance graph** method with 20 nearest neighbors. Observing the 'knee' of the curve, we identified a target range between **0.7** and **1.5** ([see Figure 22](#)). We conducted a sensitivity analysis within this interval, testing different computational algorithms (`ball_tree`, `kd_tree`, `brute`), which resulted in identical results.

The final model was instantiated with **eps = 1.4**, identifying **3** distinct clusters (excluding outliers).

3.2.2.4.3 Gaussian Mixture Model

In our **Gaussian Mixture Models (GMM)** analysis, we initially determined the optimal number of components by minimizing the **BIC** and **AIC** across a range of 1 to 10 components. The statistical 'elbow point' suggested **5** clusters ([see Figure 24](#)). To refine the model structure, we conducted a grid search evaluating different covariance types (**'full', 'tied', 'diag', 'spherical'**) for cluster counts between 1 and 6. Although 5 clusters yielded marginally better metrics, the **4-cluster** solution using a **'tied' covariance** type offered distinct advantages in terms of interpretability and business applicability.

3.2.2.5 Model Combination - SOM + K-Means

We implemented a hybrid **SOM + K-Means** model to leverage the strengths of both algorithms. By applying K-Means to the features extracted from the SOM, we identified **5 distinct clusters**.

3.2.2.6 Model Assessment

We selected **K-Means (k=4)** as the final model due to its superior business interpretability and alignment with the airline's operational structure.

While other models produced competitive statistical scores, K-Means successfully identified four distinct and actionable personas: **Loyal Regulars**, **Occasional Leisure Travelers**, **High-Frequency Commuters**, and **Ultra-Haul Groups**.

3.2.3 Seasonal Feature Clustering

3.2.3.1 Hierarchical Clustering

To determine the optimal segmentation structure, we analysed dendrograms and performed a systematic grid search across multiple configurations (Linkages: Ward, Complete, Average; Metrics: Euclidean, Manhattan, Cosine) for various cluster counts.

Based on the evaluation of metrics, the **Ward Linkage with Euclidean Distance (k=4)** emerged as the superior model, offering the most balanced and cohesive clusters.

3.2.3.2 K-Means clustering

In our analysis, we assessed **K-means clustering** by first visualizing the dispersion (inertia) across a range of 1 to 19 clusters to identify the potential elbow point. We can observe in this figure that the elbow point is clearly **k=4** (see Figure 28). To refine this selection, we evaluated the model performance metrics for cluster counts between 2 and 7. We observed that the most robust segmentation was achieved when **k=4**.

3.2.3.3 SOM clustering

In our **Self-Organizing Maps (SOM)** analysis, we first assessed training stability by monitoring the **Quantization Error** curve over **2,000** iterations. We observed that model convergence stabilized between **1000** and **1,300** iterations (see Figure 31). Based on this window, we conducted a hyperparameter optimization search, testing variations in learning rate (**0.8, 0.9, 1**), and iteration counts.

The final model was configured on a **15 X 15 grid** with a learning rate of **1** and **1,240 iterations**. This resulted in **176 clusters**, a number deemed **too high** to be actionable for high-level customer profiling

3.2.3.4 Density-Based Clustering

3.2.3.4.1 Mean-Shift Clustering

In our analysis using **Mean Shift**, a density-based algorithm that does not require specifying the number of clusters a priori, the critical hyperparameter was the bandwidth (kernel size). To determine the optimal window for density estimation, we performed a sensitivity analysis by iterating through quantiles ranging from **0.1 to 0.65** using the `estimate_bandwidth` function.

The most robust segmentation was identified at a **quantile of 0.5** (bandwidth of approx. **0.25**) resulting in a **4-cluster** solution

3.2.3.4.2 DBSCAN

For the **DBSCAN** analysis, we determined the optimal neighborhood radius (`eps`) using the **K-distance graph** method with 20 nearest neighbors. Observing the 'knee' of the curve, we identified a target range between **0.9** and **1.4** (see Figure 34). We conducted a sensitivity analysis within this interval, testing different computational algorithms (`ball_tree`, `kd_tree`, `brute`), which resulted in identical results.

The final model was instantiated with **eps = 1.1**, identifying **4** distinct clusters (excluding outliers).

3.2.3.4.3 Gaussian Mixture Model

In our **Gaussian Mixture Models (GMM)** analysis, we initially determined the optimal number of components by minimising the **BIC** and **AIC** across a range of 1 to 10 components. The statistical 'elbow point' suggested **5** clusters (see Figure 36). To refine the model structure, we conducted a grid search evaluating different covariance types ('full', 'tied', 'diag', 'spherical')

Commented [C06]: 1. Which clustering method(s) produced the most interpretable and stable results for each perspective?
2. How many clusters best represent the customer base overall, and what evidence supports this decision
3. What differentiating features emerged as most important within each perspective, and how do they complement one another in the merged solution?
4. How would you describe the final set of customer segments

for cluster counts between 2 and 6. Although 5 clusters yielded marginally better metrics, the **4-cluster** solution using a **'tied' covariance** type offered distinct advantages in terms of interpretability and business applicability.

3.2.3.5 Model Combination - SOM + K-Means

We implemented a hybrid **SOM + K-Means** model to leverage the strengths of both algorithms. By applying K-Means to the features extracted from the SOM, we identified **4 distinct clusters**, similar to K-Means.

3.2.3.6 Model Assessment

We selected **K-Means (k=4)** as the final model due to its superior business interpretability and alignment with the airline's operational structure.

While other models produced competitive statistical scores, K-Means successfully identified four distinct and actionable personas: **Year-Round**, **Summer**, **Autumn**, and **Winter travellers**.

4.0 Results & Validation

We opted for a seasonality-driven segmentation using **Hierarchical Clustering (Ward, Euclidean, k=4)** (see Figure 38) to design a precise campaign calendar. **Feature importance analysis** confirmed **Seasonal Flight Ratios** as the primary drivers—outweighing static demographics—to effectively group customers by their **temporal travel patterns** (see Figure 39). This model partitions the customer base into four actionable profiles, ordered by market share:

1. **Cluster 0 "The Routine Travelers" (69%, ~10,300 customers):** The revenue backbone is characterised by a **"flat" seasonal** profile and **consistent year-round activity**. With the highest positive **DaysActive (+0.39)** and **TotalFlights**, they represent the **Loyal Core** and necessity travellers, making them the primary target for retention strategies.
2. **Cluster 2 "The Summer Vacationers" (21.7%, ~3,200 customers):** The strategic leisure segment, defined by a sharp **SummerFlightRatio (+0.61)** and the highest **Redemption Rate (+0.47)**. **Despite shorter average flight distances (-0.29)**, their **high intensity** during active months confirms them as travellers accumulating points for regional or medium-haul family vacations.
3. **Cluster 1 "The Autumn Escapees" (5.7%, ~900 customers):** A tactical niche segment active solely in autumn (**+2.7 ratio**) with minimal engagement during the rest of the year, representing an opportunity for off-peak stimulation.
4. **Cluster 3 "The Winter Elite" (3.7%, ~600 customers):** An exclusive group of winter/holiday travellers (**+3.5 ratio**) with the highest **average flight distance (+1.2)** and a **distinctive positive Companion Ratio (+0.39)**. This identifies them as premium **long-haul tourists** travelling with **partners or family**, suitable for high-value holiday offers.

Visual validation via **Heatmap** and **Parallel Coordinates** confirms distinct, non-overlapping seasonal blocks and significant deviations from the mean (see Figure 40 and 41).

5.0 Strategic Recommendations

Based on our analysis, the behavioural patterns identified suggest a shift towards **Calendar-Driven Marketing**. We recommend implementing the following targeted strategies:

Commented [C07]: Analyze clustering performance through technical metrics and statistical significance testing. Interpret business meaning of discovered segments with customer profiles and behavioral characteristics.

Commented [C08]: Develop actionable strategies for each customer segment, including marketing approaches, service customization, and loyalty program optimization for AIAI.

5.1 Retention Strategy for "Routine Travelers" (Cluster 0)

This segment's primary objective is **retention and churn prevention**.

1. **Operational Focus:** Focus on reducing friction across the trip. This group values efficiency, consistency, and reliability over seasonal offers.
2. **Loyalty Engagement:** Offer perks that improve the travel experience year-round (e.g., fast-track security, lounge access trials, or double points on business routes).
3. **Brand Relationship:** Besides the avoidance of inconveniences, the best way to keep customers happy is by having them build a relationship with the brand. Use surprise gifts and a monthly newsletter to keep the brand top-of-mind.

5.2 Monetisation Strategy for "Summer Vacationers" (Cluster 2)

1. **Early campaign:** Shift marketing for this group to **March/April** to capture their booking intent before competitors do, and before the customers' vacations are booked.
2. **Upsell:** Since these are likely family or leisure trips, aggressively market "add-ons" such as extra baggage, seat selection, and travel insurance during the booking flow.

5.3 Stimulation Strategy for Niche Segments (Clusters 1 & 3)

1. **The Autumn Push (Cluster 1):** Launch tactical "City Break" campaigns targeting this specific group that is mostly comprised of regular travellers. Focus on filling empty seats by using price-led promotions to stimulate demand for October/November flights.
2. **The Winter Premium (Cluster 3):** Focus on **high-value proposition marketing**. Target this group in October/November with exotic/ski destinations. People here tend to be willing to travel longer distances so the focus should be on providing the best experience. These are also times to upsell "add-ons" just like with Summer Vacationers.

6.0 Conclusion

This project confirms that **seasonality, not demographics, is the primary driver of passenger behaviour**. By validating that 31% of the base belongs to high-value seasonal niches while 69% are steady "Routine Travelers," we have established the case for **Calendar-Driven Marketing**. This shift allows AIAI to align spending with specific travel windows, minimising waste and maximising customer lifetime value.

Limitations of our analysis include that the current model relies on historical snapshots and excludes inactive profiles to ensure stability. Consequently, the massive "Routine" segment (69%) remains too broad, obscuring critical nuances like price sensitivity versus route convenience.

To evolve this framework into a competitive advantage, we recommend the following three immediate actions:

1. **Micro-Segment the Core:** Drill down into the "Routine" group to uncover hidden sub-behaviours and ancillary revenue opportunities.
2. **Predict Churn:** Move from descriptive analysis to predictive modelling to forecast risk and booking intent before it happens.
3. **Automate Recommendations:** Integrate these seasonal profiles into a live engine that delivers personalised destination offers in real-time.

Commented [C09]: Integrate technical findings with business strategy, address methodology limitations, and outline future research directions for AIAI's segmentation approach.

7.0 Annexes

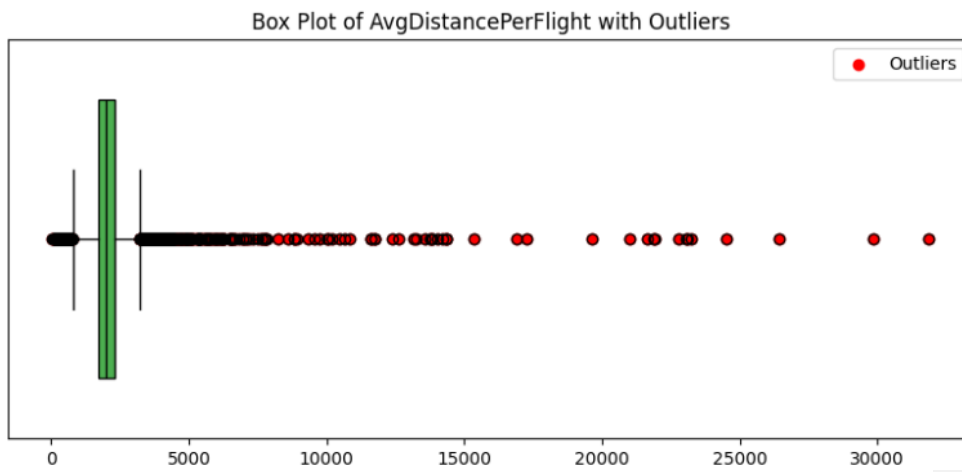


Figure 1 – AvgDistancePerFlight Outliers

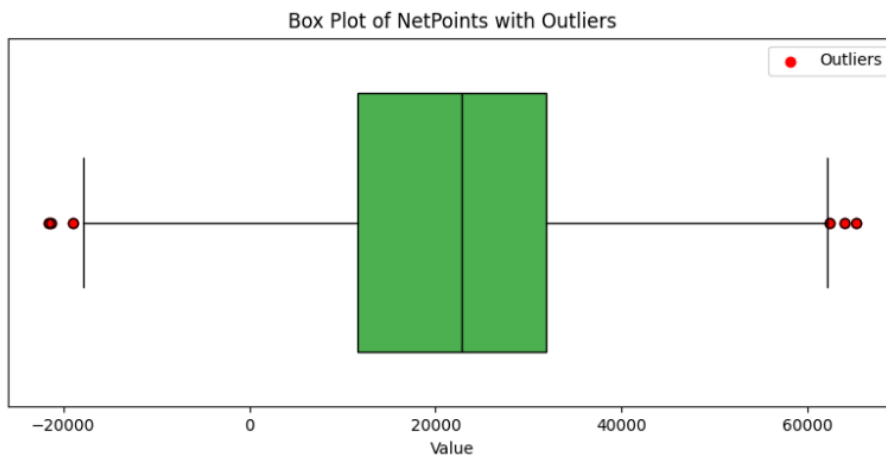


Figure 2 – NetPoints Outliers

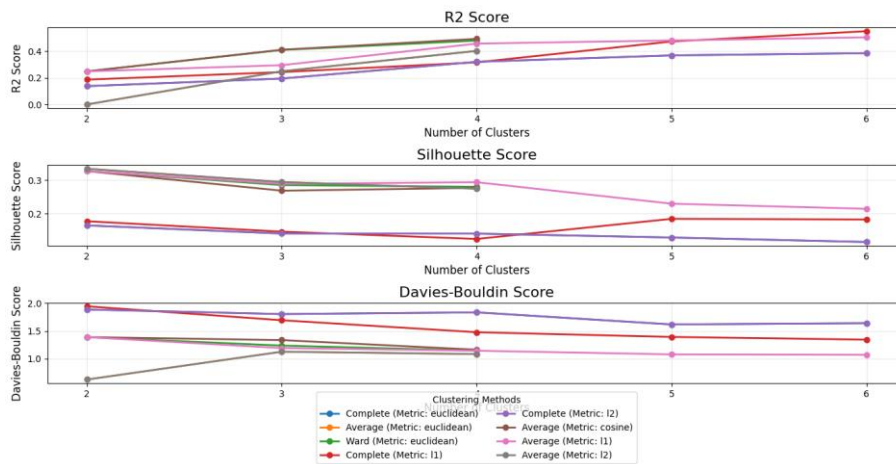


Figure 3 - Value Perspective Hierarchical Merics

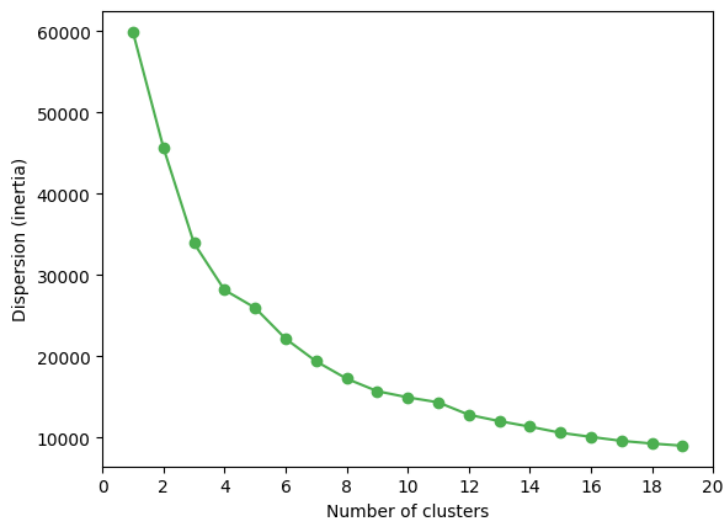


Figure 4 - Value Perspective K-Means Graph

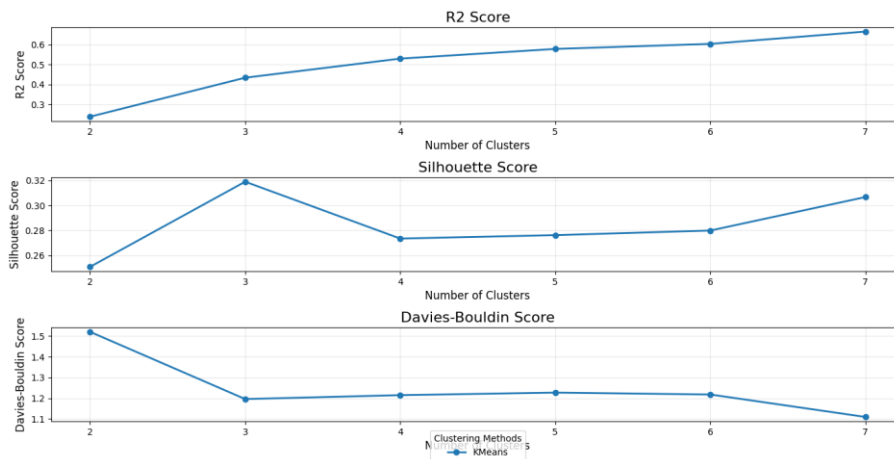


Figure 5 - Value Perspective KMeans Metrics

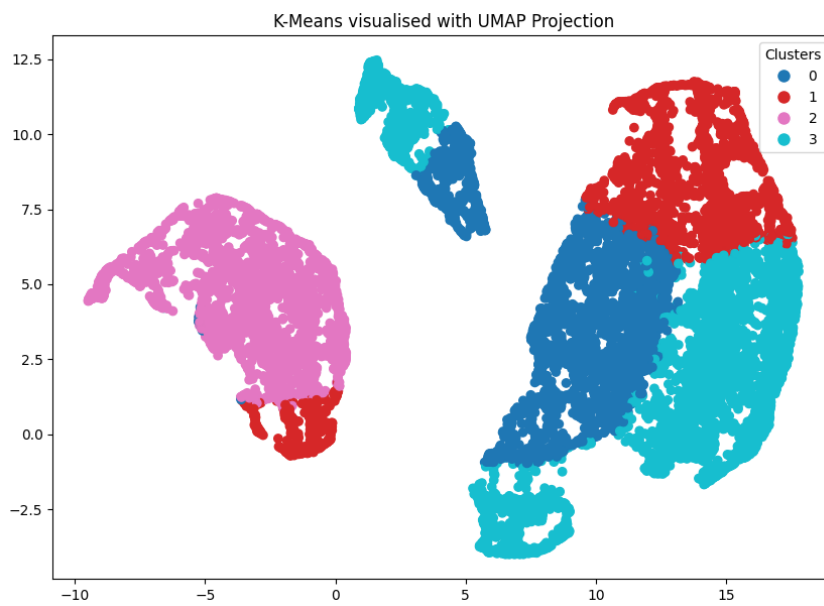


Figure 6 - Value Perspective K-Means UMAP (best model)

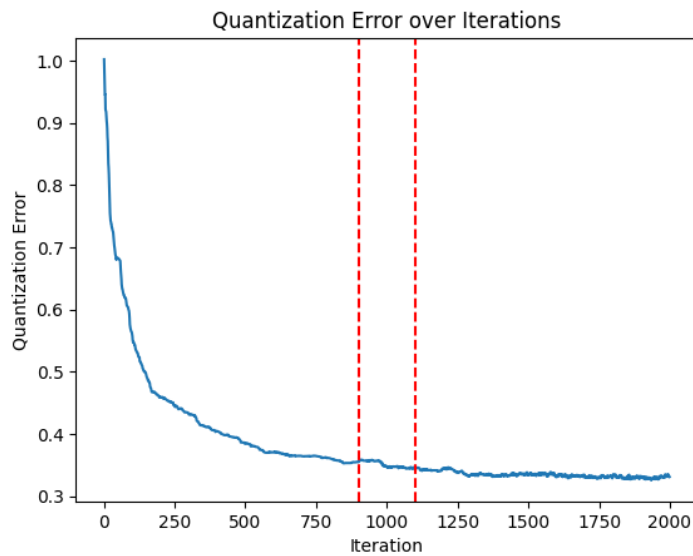


Figure 7. Value Perspective SOM Graph

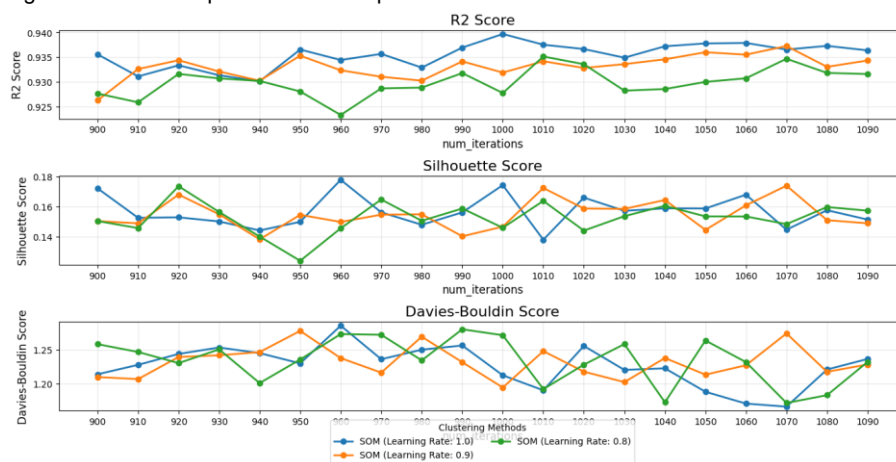


Figure 8. Value Perspective SOM Metrics

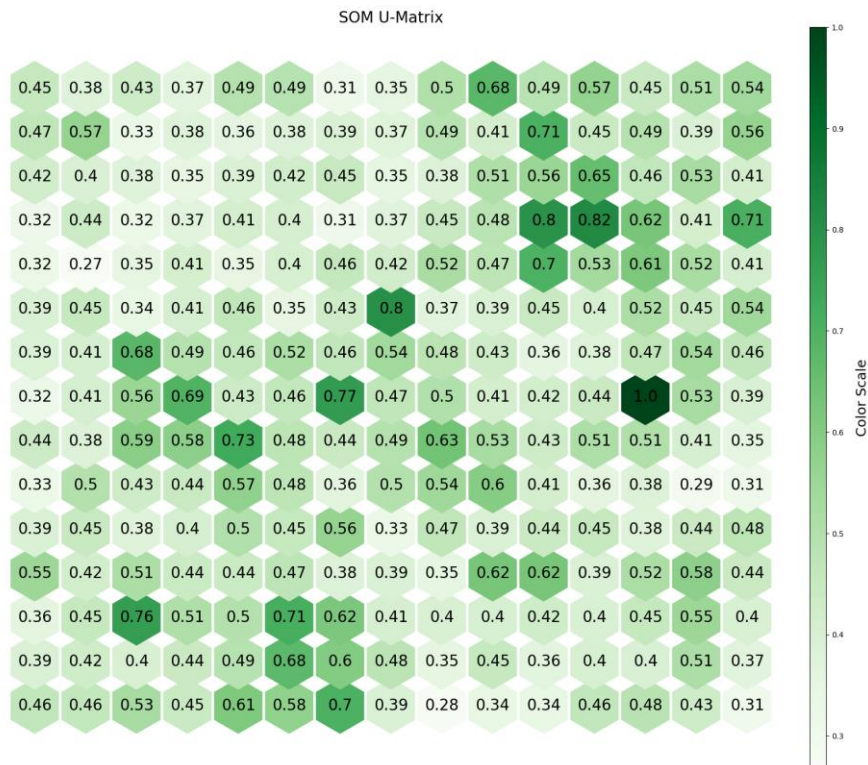


Figure 9. Value Perspective SOM U-Matrix

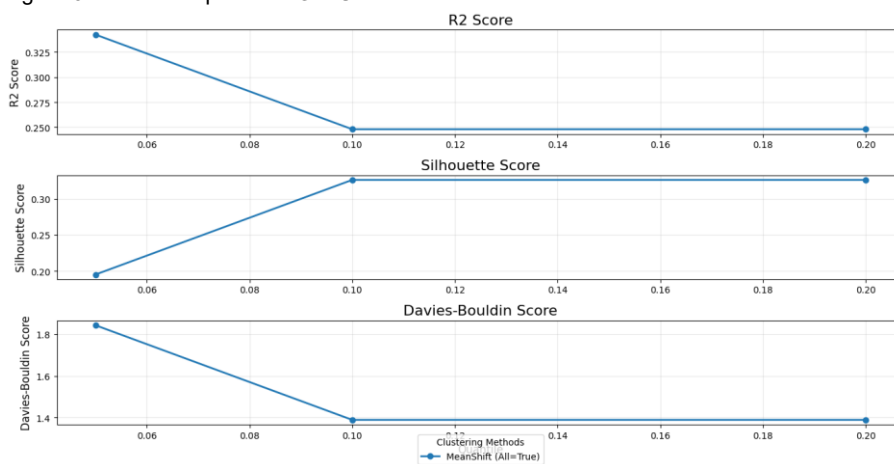


Figure 10. Value Perspective Meanshift Metrics

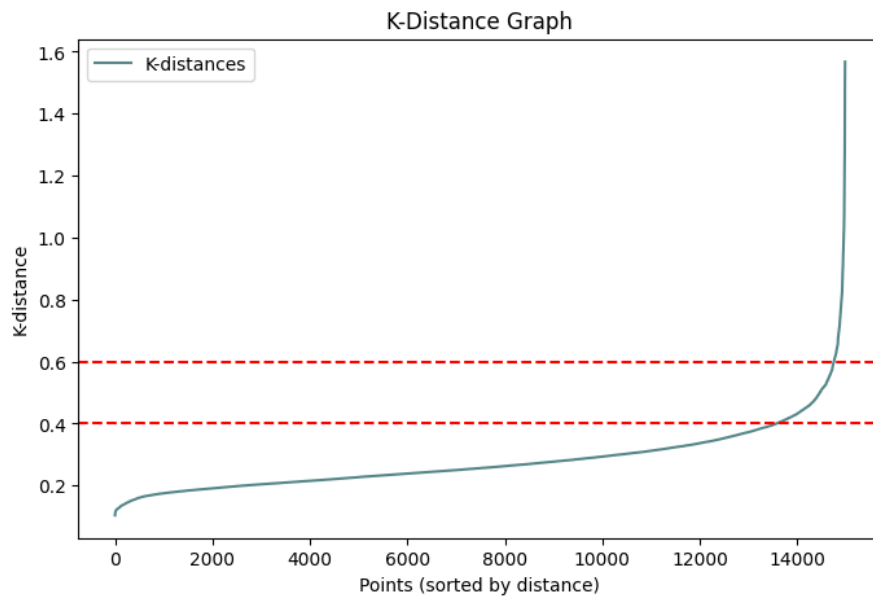


Figure 11 - Value Perspective DBSCAN Graph

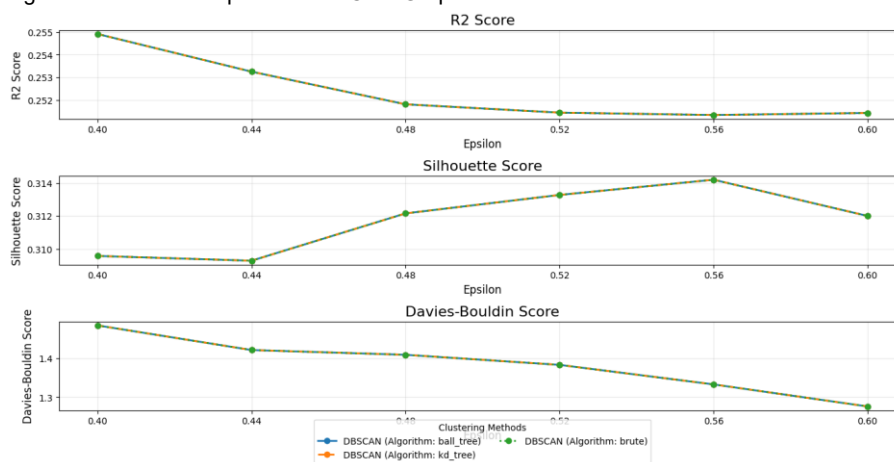


Figure 12. Value Perspective DBSCAN Metrics

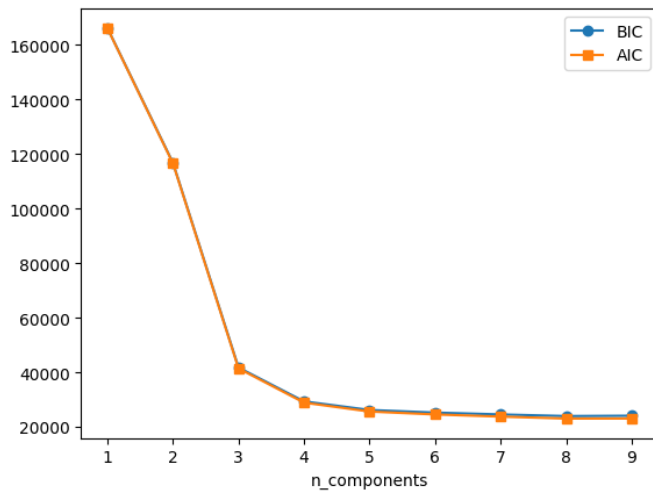


Figure 13 – Value Perspective GMM Graph

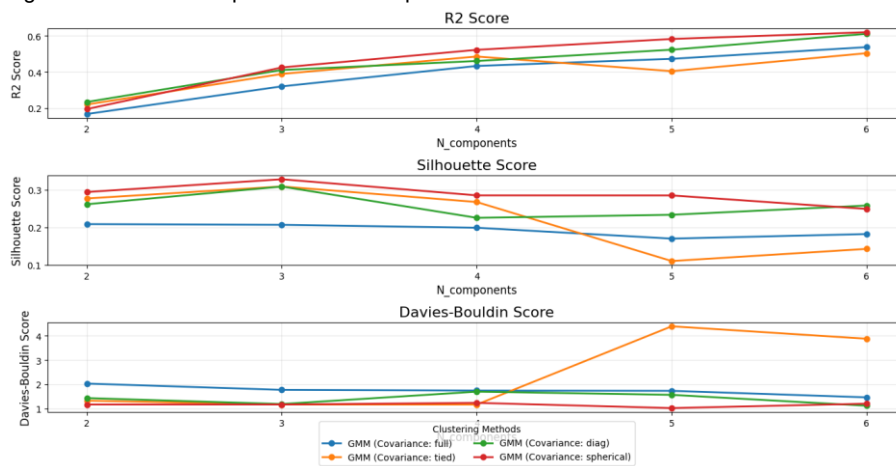


Figure 14 – Value Perspective GMM Metrics

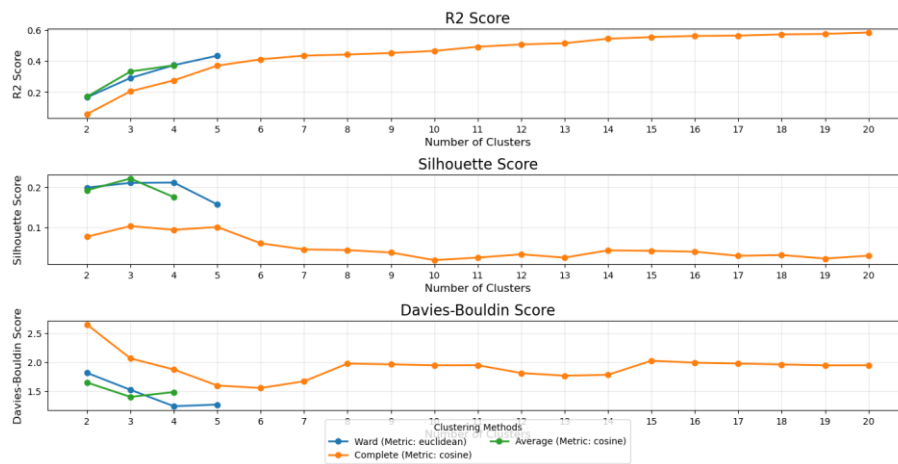


Fig 15 - Frequency & Volume Perspective Hierarchical metrics

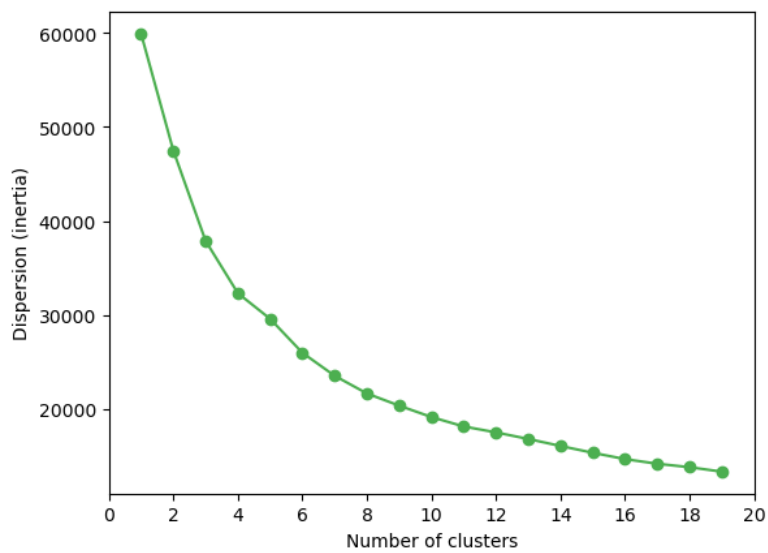


Figure 16 - Frequency & Volume Perspective KMeans Graph

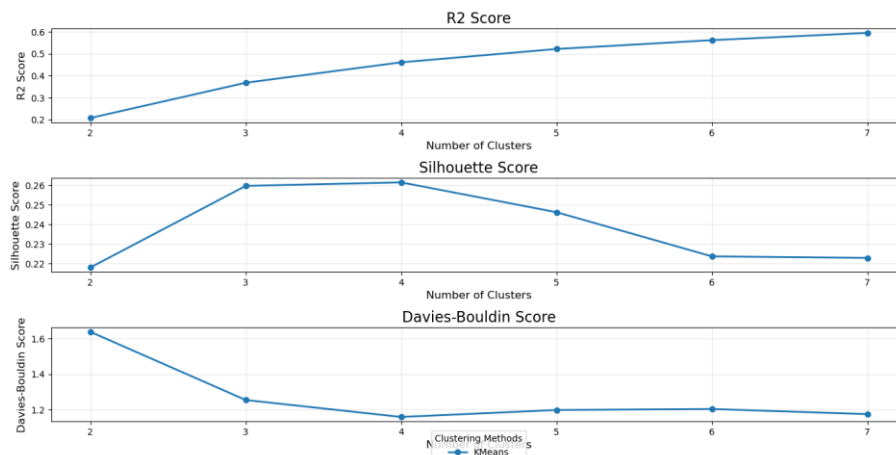


Fig 17 - Frequency & Volume Perspective K-means metrics



Figure 18 - Frequency & Volume Perspective SOM Graph

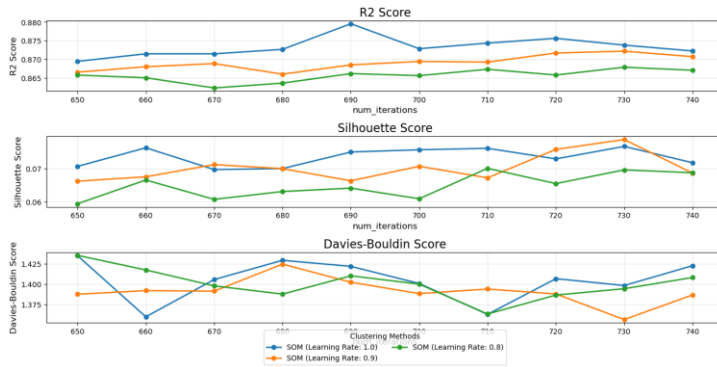


Fig 19 - Frequency & Volume Perspective SOM metrics

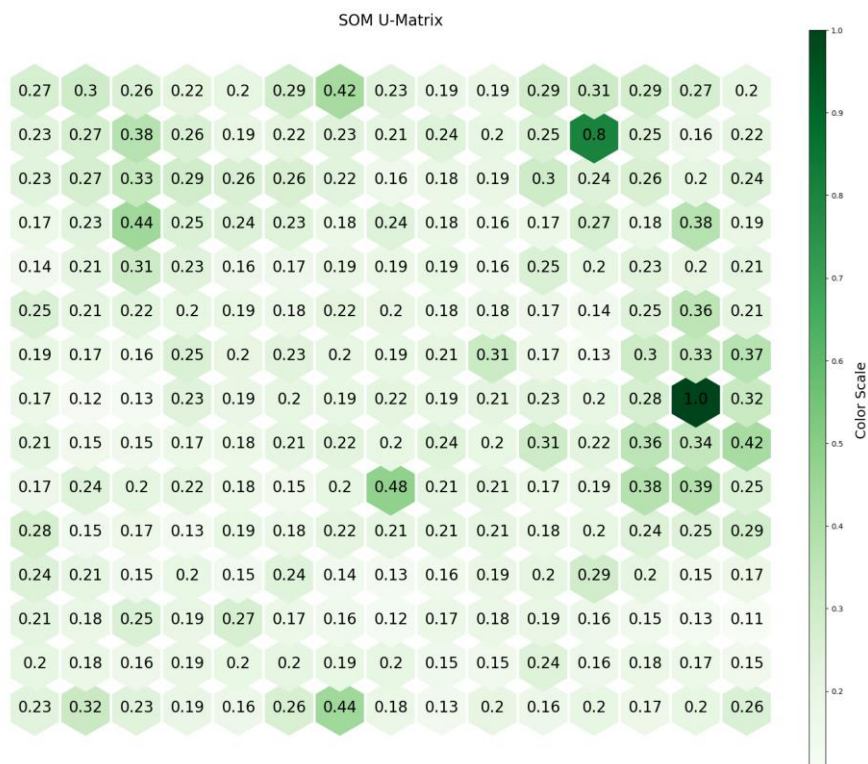


Fig 20- Frequency & Volume Perspective SOM U-metrics

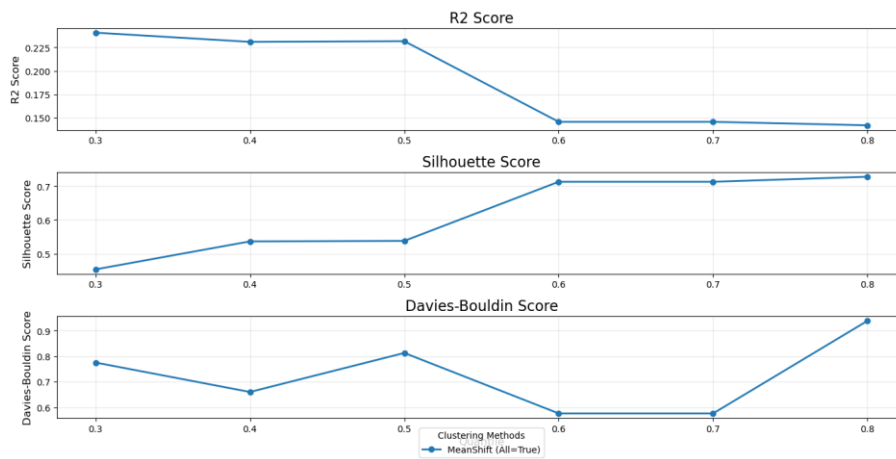


Fig 21 - Frequency & Volume Perspective MeanShift metrics

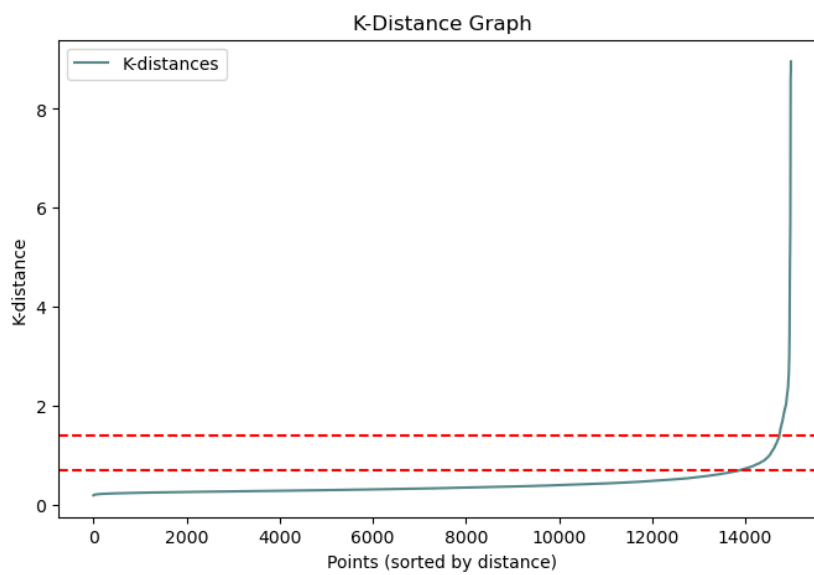


Figure 22 - Frequency & Volume Perspective DBSCAN Graph

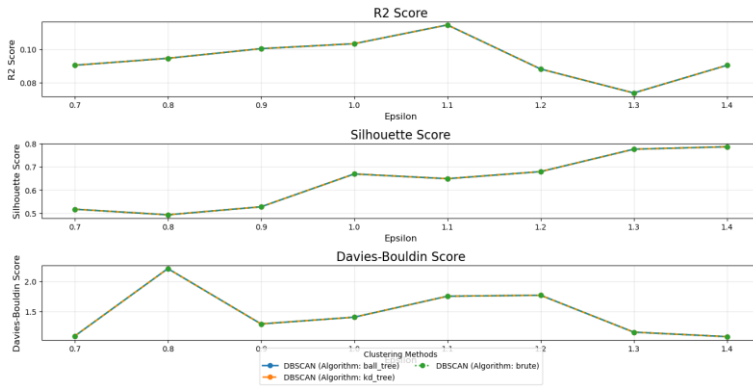


Figure 23 - Frequency & Volume Perspective DBSCAN metrics

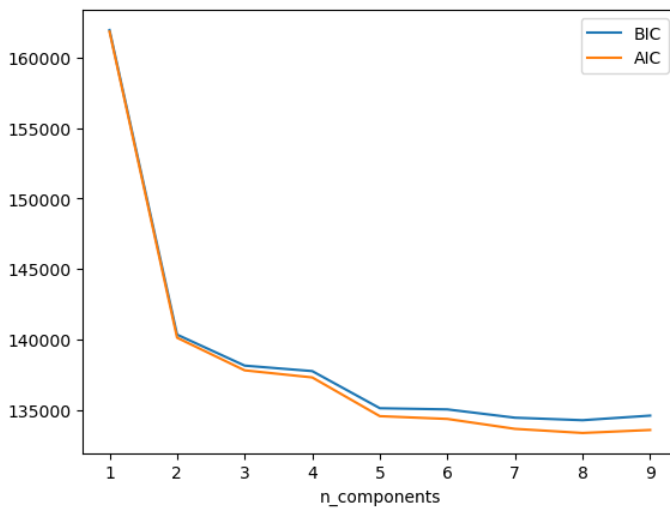


Figure 24 - Frequency & Volume Perspective GMM Graph

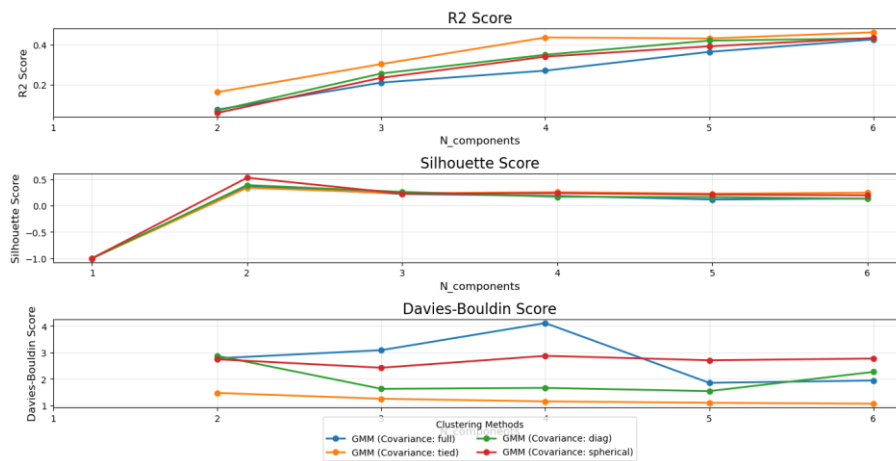


Figure 25 - Frequency & Volume Perspective GMM metrics

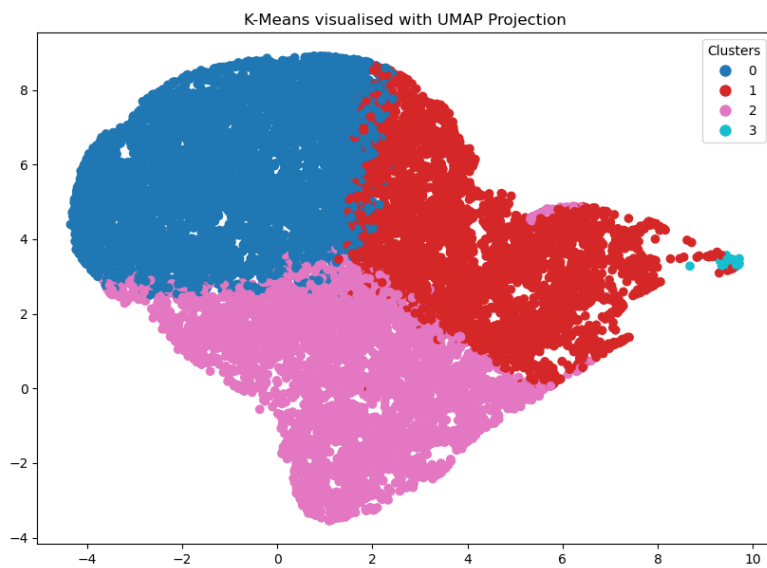


Figure 26 - Frequency & Volume Perspective K-Means UMAP (best model)

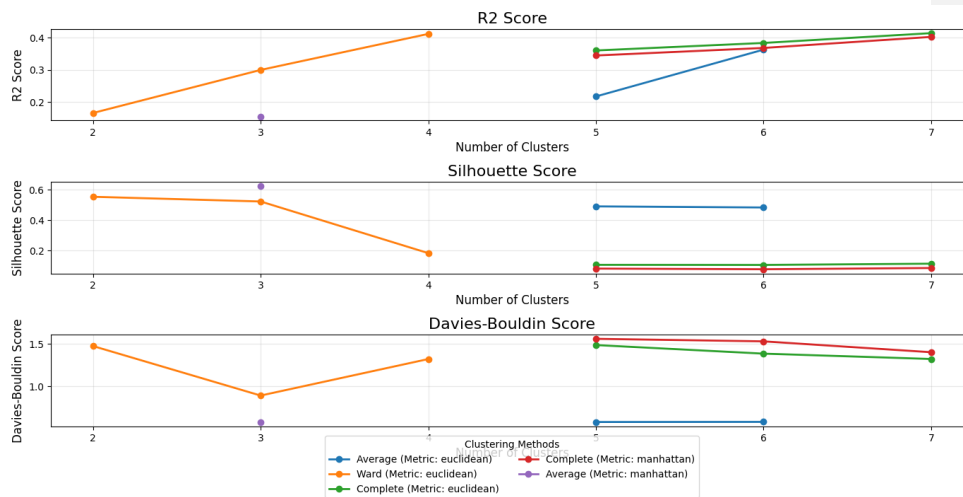


Figure 27 – Seasonal Perspective Hierarchical Metrics

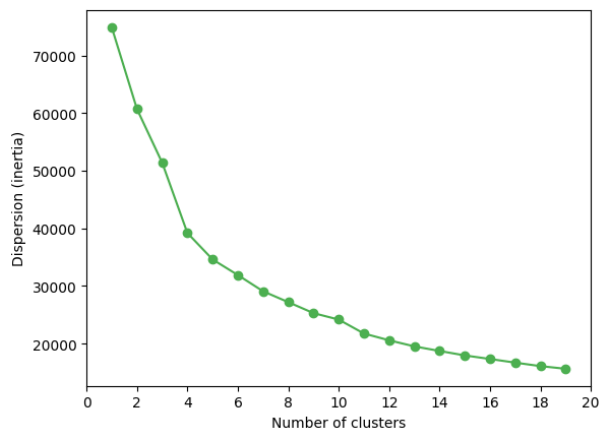


Figure 28 – Seasonal Perspective K-Means Graph

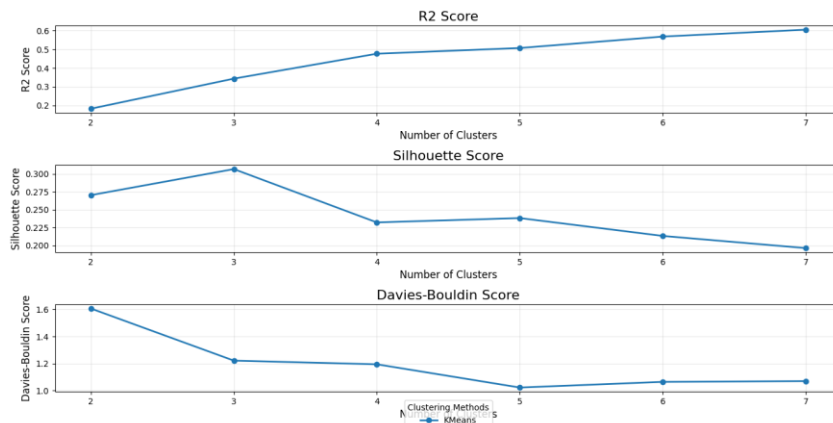


Figure 29 – Seasonal Perspective K-Means Metrics

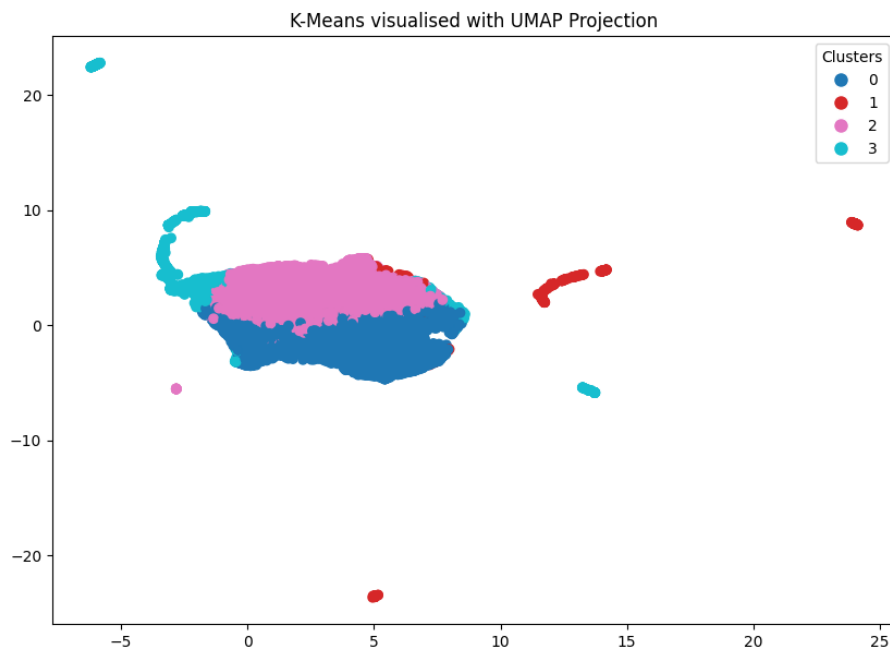


Figure 30 – Seasonal Perspective K-Means UMAP (best model)

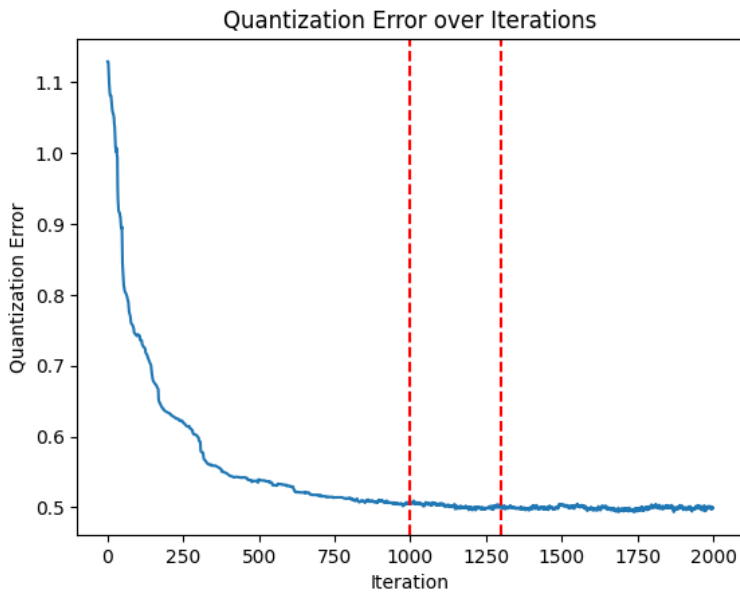


Figure 31 – Seasonal Perspective SOM Graph
SOM U-Matrix

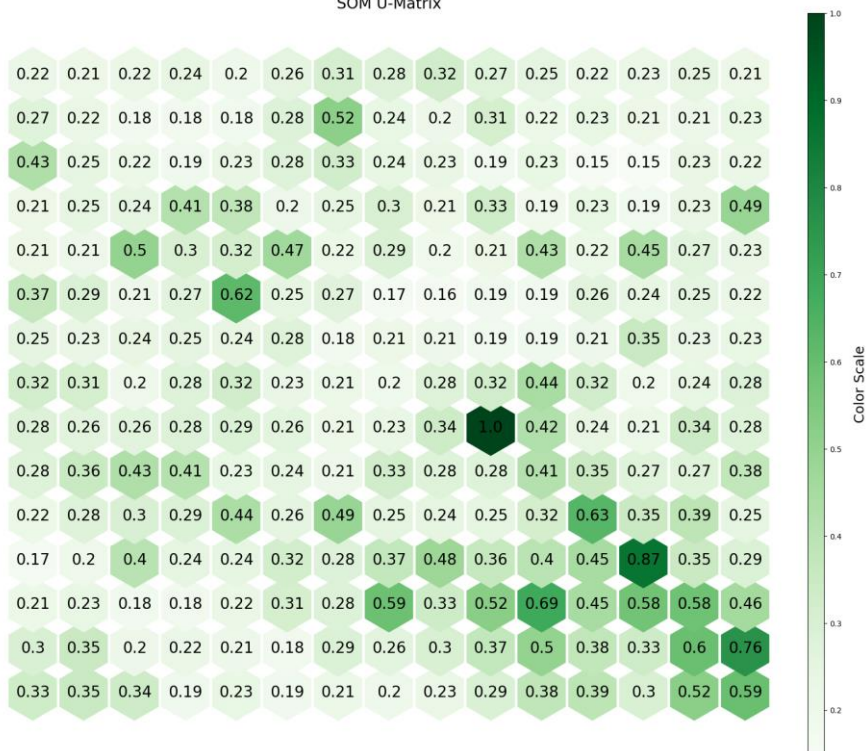


Figure 32 – Seasonal Perspective SOM U-Matrix

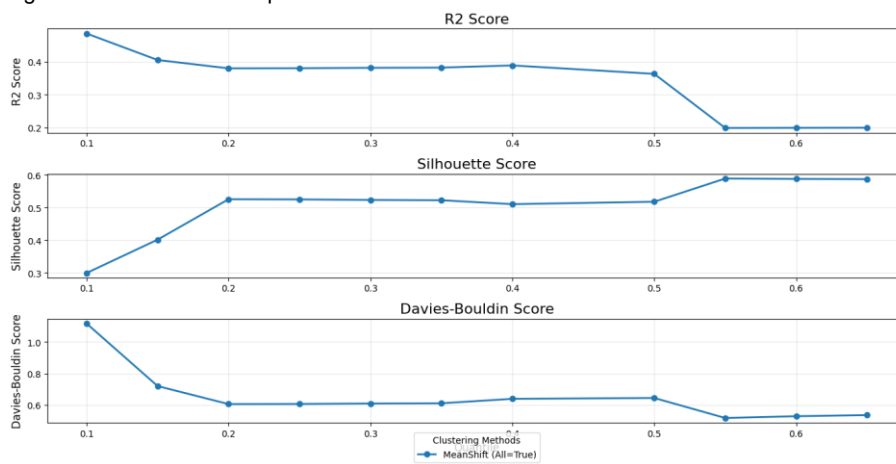


Figure 33 – Seasonal Perspective MeanShift Metrics

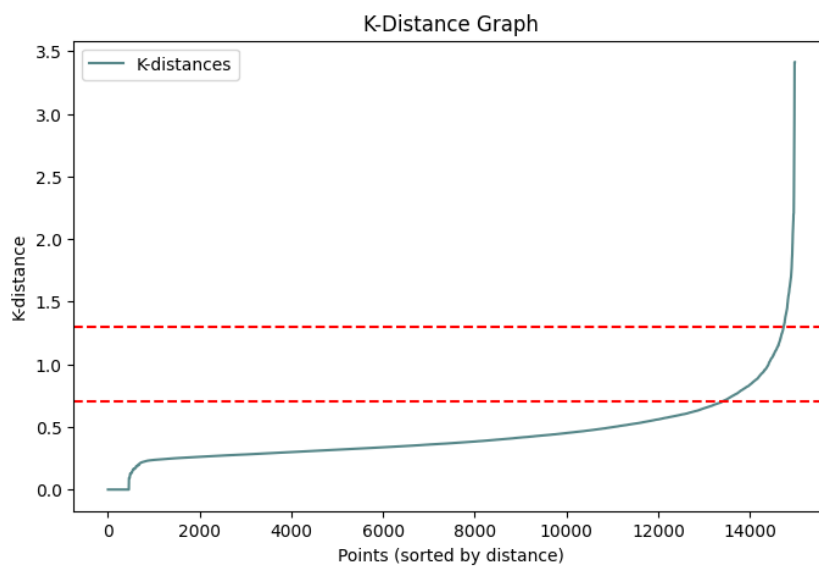


Figure 34 – Seasonal Perspective DBSCAN Graph



Figure 35 – Seasonal Perspective DBSCAN Metrics

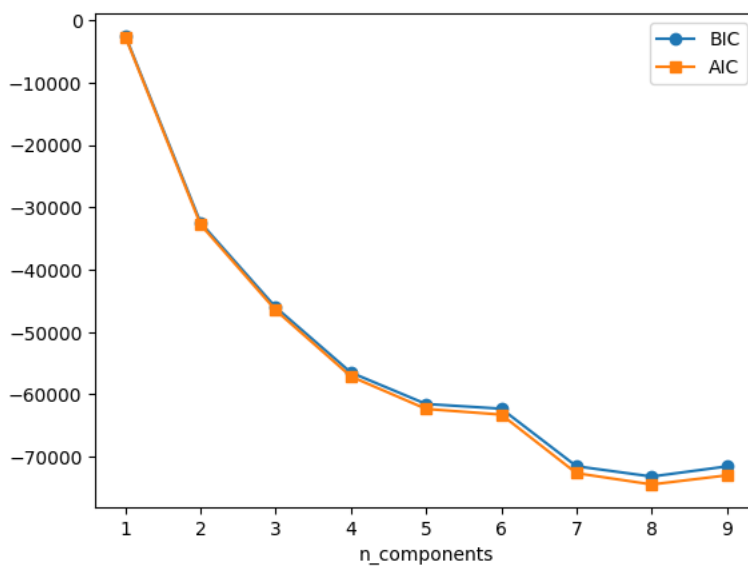


Figure 36 – Seasonal Perspective GMM Graph

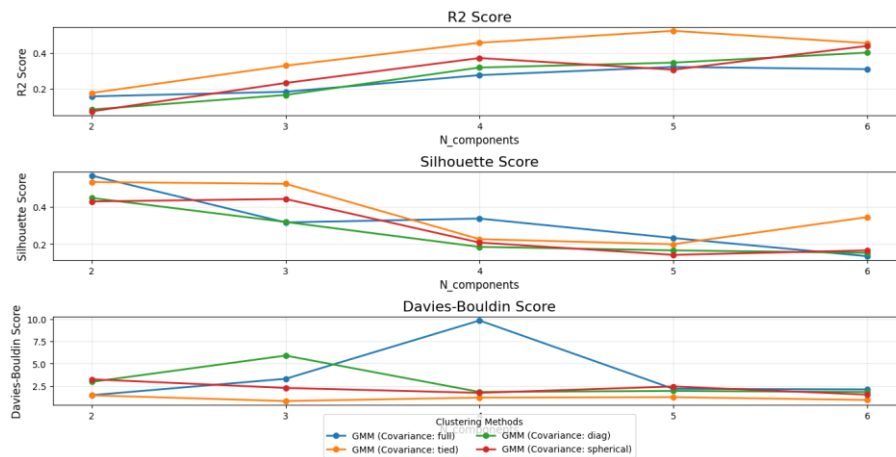


Figure 37 – Seasonal Perspective GMM Metrics

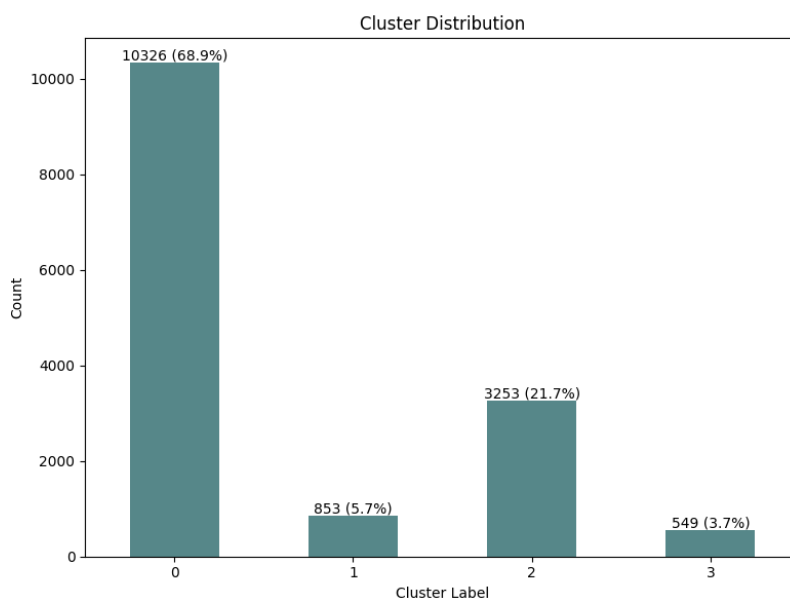


Figure 38 – Merged Cluster Label Countings

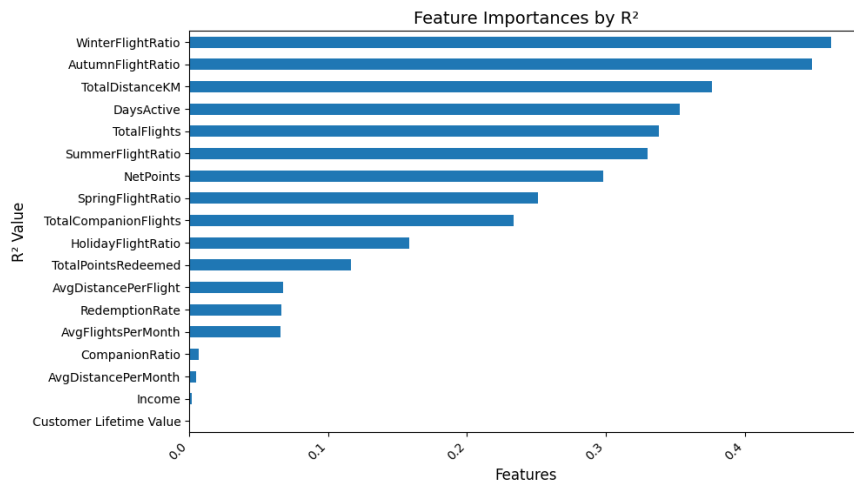


Figure 39 – Merged Feature Importance by R^2

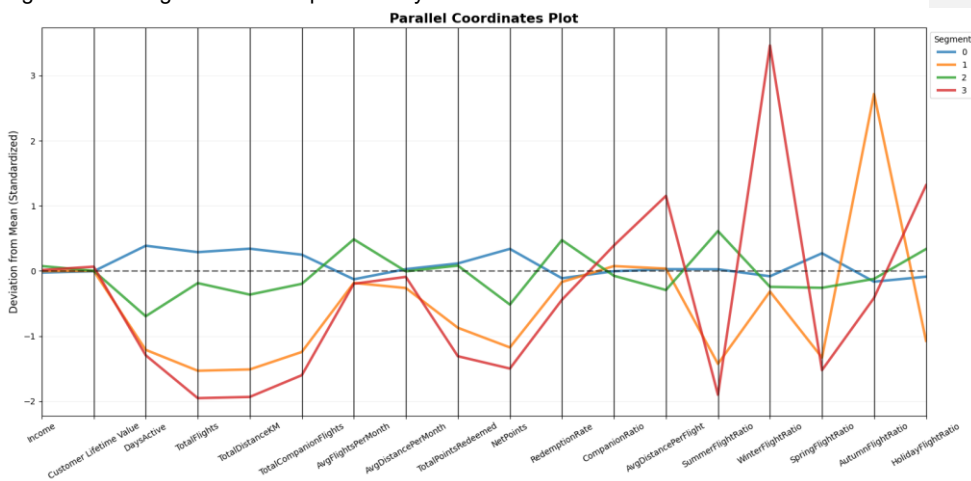


Figure 40 – Merged Parallel Coordinates

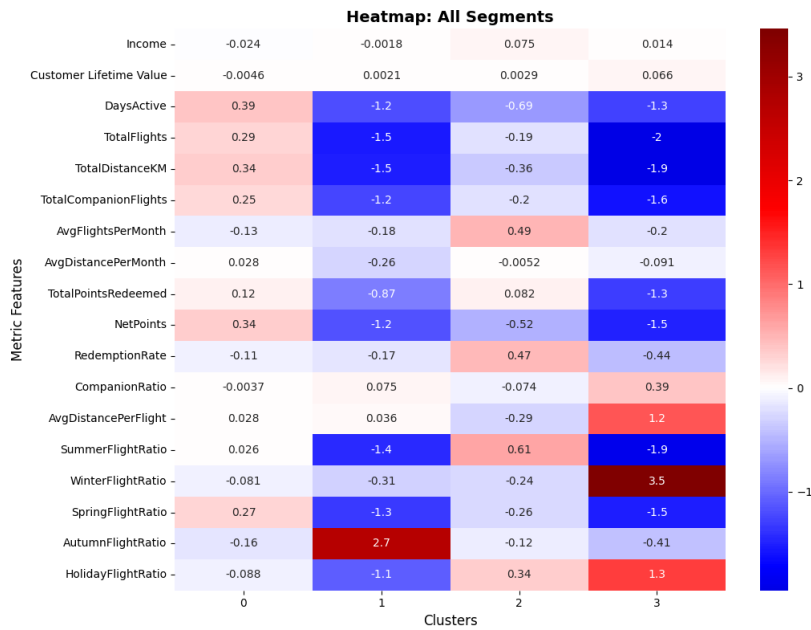


Figure 41 – Merged Heatmap

New Variable	Description	Variable(s) Used
ISActive	customers who have not cancelled or whose enrollment date is greater than the last cancellation date	EnrollmentDateOpening CancellationDate
DaysActive	Number of days the customer has been active since the last loyalty subscription date	ISActive EnrollmentDateOpening CancellationDate
HasFlown	Number of customers who have at least one flight	NumFlights Loyalty#
IsValuable (CLV Outlier)	Customers with high lifetime values	Customer_Lifetime_Value
IsReEnrolled	Customers enrolled on the Loyalty Program	EnrollmentDateOpening CancellationDate
LoyaltyProfile	Customers overall loyalty behaviour	TotalPoitsAccumulated RedemptionRate
TotalFlights	Total number of flights ever taken.	NumFlights

TotalDistanceKM	Total distance travelled.	DistanceKM
TotalCompanionFlights	Total flights taken with other people.	NumFlightsWithCompanions
AvgFlightsPerMonth	Average flights per monthly	NumFlights
AvgDistancePerMonth	Average distance flown per monthly	DistanceKM
TotalPointsAccumulated	Total points earned by the customer.	PointsAccumulated
TotalPointsRedeemed	Total number of points redeemed for rewards.	PointsRedeemed
NetPoints	Difference between earned and redeemed points	TotalPointsRedeemed TotalPointsAccumulated
Redemption Rate	How many earned points were utilised	TotalPointsRedeemed TotalPointsAccumulated
CompanionRatio	Proportion of flights taken with companions	TotalCompanionFlights TotalFlights'
Average Distance per Flight	Typical flight length	TotalDistanceKM TotalFlights
SummerFlightsRatio	Travels during the summer season	TotalFlights YearMonthDate
WinterFlightsRatio	Travels during the winter season	TotalFlights YearMonthDate
SpringFlightsRatio	Travels during the spring season	TotalFlights YearMonthDate
AutumnFlightsRatio	Travels during the autumn season	TotalFlights YearMonthDate
HolidayFlightRatio	Travels during the holiday season	TotalFlights YearMonthDate
TravellerType	The distinctions of Travellers based on	SummerFlightsRatio WinterFlightsRatio SpringFlightsRatio AutumnFlightsRatio HolidayFlightRatio CompanionRatio

Table 1

Annex AI Usage Statement

AI tools were used for coding syntax assistance (ChatGPT and Gemini for pandas operations, matplotlibs and graph plotting) and report proofreading (Grammarly and Quillbot for grammar checks and Gemini for removing redundancy in language). All analytical insights, business interpretations, and strategic recommendations represent original group analysis and thinking.

Contribution Statement

Ricardo Isidro (Student ID: 20250374)

- Business Understanding
- Feature Engineering Ideation
- Report proofreading
- Data Visualisations
- Feature Transformation
- Merging of the Clusters
- Advanced Profiling & Business Persona Definition
- Feature Importance Analysis

Rodrigo Santos (Student ID: 20250387)

- Preprocessing Pipeline
- Feature Engineering & Transformation
- Outlier Detection & Handling
- Independent Segmentation (Value, Frequency & Volume, Seasonality)
- Comparative Algorithm Analysis (Hierarchical, K-Means, SOM, Density-Based)
- Hyperparameter Tuning & Model Selection
- Clustering Implementation (Hierarchical Ward Method)
- Advanced Profiling & Business Persona Definition
- Feature Importance Analysis
- Technical Analysis
- Strategic Reporting
- Project Assurance

Rodrigo Teixeira (Student ID: 20250393)

- Business Understanding
- Incoherence Ideation
- Feature Engineering Ideation
- Feature Engineering & Transformation
- Outlier Detection & Handling
- Independent Segmentation (Value, Frequency & Volume, Seasonality)
- Comparative Algorithm Analysis (Hierarchical, K-Means, SOM, Density-Based)
- Hyperparameter Tuning & Model Selection
- Technical Analysis
- Strategic Reporting
-

Christian Chukwuebuka Ozougwu (Student ID: 20250447)

- Business Understanding
- Report writing and coordination
- Infographic Section and Poster Design
- Feature Engineering Ideation
- Incoherence Ideation
- Data Visualisations
- Clustering application to Segment
- Development of Presentation Slides
- Video Presentation Coordination

All members contributed to collaborative discussions and ideation.

Responsibility Statement

We, the group members listed above, certify that this report represents our original analytical work and interpretations. While AI tools were used as specified above, all insights, conclusions, and recommendations are the result of our independent analysis and critical thinking. We take full responsibility for the accuracy and quality of this submission.